INFERENCE FROM STOCHASTIC BIOLOGICAL SIMULATORS USING NEURAL POSTERIOR ESTIMATION

A PREPRINT

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ABSTRACT

Parameter inference is essential for translating sparse, noisy biological measurements into quantitative, mechanistic insights. While inference methods are well-established for deterministic models, many biological applications require stochastic approaches to capture inherent randomness. Agent-based random walk models are particularly effective for simulating spatiotemporal processes like ecological dynamics, molecular transport, and multicellular phenomena. However, inference for these models is challenging due to intractable likelihoods, which often necessitate surrogate approximations, pre-specified noise models, or likelihood-free algorithms like Approximate Bayesian Computation (ABC) that are sensitive to the choice of distance metrics and tolerances. We present a machine learning workflow using Neural Posterior Estimation (NPE) that overcomes these challenges. By learning the posterior distribution directly from simulations, NPE enables amortized inference for near-instantaneous posterior estimation. Our pipeline delivers well-calibrated posteriors and uncertainty-aware forecasts through posterior predictive sampling. It also includes principled diagnostics via simulation-based calibration and coverage assessment, all without committing to a potentially misspecified likelihood or a continuum surrogate. We replicate previous baselines (ABC and likelihood-based methods) and demonstrate that NPE achieves improved calibration and predictive validity on both synthetic and experimental datasets with competitive simulation budgets. To facilitate adoption and extension, we release an open, reproducible implementation with configuration files, plots, and artifact logs, compatible with command-line and SLURM execution.

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1 Introduction

Parameter inference is the critical link between mathematical models and biological understanding, enabling quantitative insights from noisy experimental data. Inference methods provide researchers with a powerful toolkit to generate predictions about system behavior, estimate unknown parameters, test competing hypotheses, and rigorously quantify uncertainty. These applications are critical across a wide range of fields, including epidemiology [Dunson, 2005, Semochkina and Walsh, 2025], ecology [Qian et al., 2009, Rosen et al., 2001], cellular biology [Vallejos et al., 2015], gene regulatory networks [Liu et al., 2016, Wang et al., 2019], systems biology [Rabinowitz and Sridhar, 2013] phylogenetics [Nascimento et al., 2017], and neuroscience [Coventry and Bartlett, 2024].

For deterministic continuum models described by ordinary and partial differential equations, inference methodologies are mature and well-established. Both Bayesian [Bernardo and Smith, 1994, Gelman et al., 1995] and frequentist [Cox and Hinkley, 1974, Casella and Berger, 1990] approaches provide robust foundations for parameter estimation, model validation, and uncertainty quantification. The sophistication of these inference techniques has enabled reliable parameter estimation and prediction across numerous scientific disciplines including climate science [Vrugt et al., 2014],

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cosmology [Planck Collaboration et al., 2020], fluid dynamics [Cheung et al., 2011] gravitatinal wave astrophysics [Veitch et al., 2015].

Stochastic biological models, however, present fundamental inferential challenges that resist these established approaches. Agent-based models and discrete stochastic processes, while capturing essential biological variability and heterogeneity, typically possess intractable likelihoods that preclude standard statistical methods. Experimental protocols exhibit substantial variation, datasets are frequently sparse and noisy, and the stochastic nature of biological processes introduces complex dependencies that deterministic frameworks cannot adequately represent. These characteristics create a disconnect between the sophisticated stochastic models needed to represent biological reality and the inference tools available to calibrate them from data.

Several recent developments have attempted to address these inferential challenges while preserving the biological realism of stochastic models [Carr et al., 2021, Simpson et al., 2022, Simpson and Plank, 2025], typically focusing on inference, identifiability, and prediction for stochastic agent-based models of migration and proliferation. These approaches often use experimental data such as scratch (barrier) assay experiments to calibrate high-fidelity discrete random-walk models, highlighting applications where stochasticity is visible and decision-relevant. At the same time, these successes have clarified several canonical challenges that have historically limited the broader application of stochastic models. While these approaches demonstrate the potential of sophisticated inference frameworks, they also reveal fundamental limitations that must be addressed to fully realize the promise of stochastic biological modeling. These canonical challenges can be summarized as follows.

1. **Intractable likelihoods:** The likelihood function for most agent-based models is intractable, precluding the use of standard methods like Maximum Likelihood Estimation or many Markov chain Monte Carlo (MCMC) algorithms.
2. **Surrogate-model error:** A common strategy to make inference tractable is to replace the computationally expensive stochastic simulator with a cheap surrogate, typically a deterministic Partial Differential Equation (PDE) describing mean-field behavior. While this enables efficient MCMC or MLE techniques, the resulting inference is conditional on the surrogate being a faithful approximation, an assumption that, if violated, can lead to biased parameter estimates and inaccurate predictions.
3. **Noise-model misspecification:** The use of a deterministic surrogate creates the problem of specifying a noise model to connect its clean output to noisy data. A misspecified choice can lead to non-physical predictions and poorly calibrated uncertainties.
4. **Likelihood-free inefficiency:** Classical likelihood-free alternatives like Approximate Bayesian Computation (ABC) are notoriously computationally expensive and sensitive to the choice of summary statistics, distance metrics, and tolerance thresholds.
5. **Information loss from summary statistics:** Classical methods often require reducing complex, high-dimensional data (e.g., the full 2D spatial arrangement of cells) to lower-dimensional summary statistics (e.g., a 1D cell density profile). This step risks a significant loss of information, and the choice of statistics can heavily bias the inference.

In this paper, we propose that these fundamental challenges can be addressed in a unified manner using Neural Posterior Estimation (NPE), a modern simulation-based inference (SBI) paradigm. NPE leverages flexible neural density estimators to learn the posterior distribution $p(\theta \mid x)$ directly from simulator-generated (θ, x) pairs. This approach offers direct remedies to each of the aforementioned challenges.

1. It is likelihood-free, circumventing intractability.
2. It learns directly from the high-fidelity simulator, eliminating surrogate model bias and removing the PDE as a source of approximation error.
3. It implicitly learns the full, complex, stochastic relationship between parameters and data, obviating the need for an explicit noise model.
4. It can ingest high-dimensional data directly (e.g., raw 2D images), preserving information and avoiding manual selection of summary statistics.
5. It provides amortized inference: after an initial, one-time training phase, it can generate posterior estimates for new observations almost instantaneously, overcoming the high per-inference cost of methods like ABC.

The central thesis of this work is that NPE provides a more robust, efficient, and direct framework for parameter inference in stochastic biological models than the classical methods reviewed in Simpson and Plank [2025] (*S&P2025* hereafter). We demonstrate this by applying a state-of-the-art Sequential NPE (SNPE) workflow to the full suite of random walk models from their study. Building on their modelling program, our contributions are to: (i) introduce an amortized, simulator-native NPE pipeline for the migration and proliferation random-walk model; (ii) demonstrate that this NPE pipeline circumvents ABC tolerance tuning and surrogate-model bias while improving calibration and coverage; (iii) introduce a novel inference strategy that uses a Convolutional Neural Network (CNN) to learn directly from the raw 2D spatial data generated by the simulator. This approach avoids the information loss inherent in using 1D summary statistics (such as column counts), leading to more precise and reliable posterior estimates; (iv) Release an open, reproducible implementation with configuration files, plots, and posterior-predictive utilities to facilitate adoption and extension. While we use scratch assay experiments as our primary motivating application, the inference framework developed here is applicable to the broader class of spatially-explicit stochastic models commonly used throughout biology and ecology.

The remainder of this paper is structured as follows. Section 2 reviews the stochastic models and classical inference paradigms and reproduces s subset of the core results of *S&P2025*. Section 3 details our NPE methodology. Section 4 presents our results, directly comparing the performance of NPE against the classical methods on the *S&P2025* benchmarks. Finally, Section 5 discusses the broader implications of our work for the future of simulation-based inference in biology.

2 Classical Approaches to Inference in Stochastic Biological Models

Our work builds upon the class of stochastic models and inference benchmarks established in *S&P2025* . Before introducing our NPE framework (Section 3), we briefly review the classical inference approaches for stochastic agent-based models, as exemplified by S&P2025's comprehensive treatment. These methods have been the workhorses of parameter inference in mathematical biology, each with distinct strengths and limitations that motivate our proposed alternative.

2.1 Approximate Bayesian Computation

ABC [Sunnåker et al., 2013, Sisson et al., 2018] represents the most direct approach to likelihood-free inference for stochastic models. As demonstrated in *S&P2025* (Section 3.1), ABC rejection sampling proceeds by: (i) sampling parameters θ from a prior distribution; (ii) simulating data $N^{(j)}$ from the stochastic model; (iii) computing a discrepancy measure $\rho(y^{\text{obs}}, N^{(j)})$ between observed and simulated data; and (iv) retaining parameters that produce sufficiently small discrepancies.

While conceptually straightforward, ABC faces well-documented challenges. The choice of tolerance threshold involves a bias-variance trade-off: small tolerances yield accurate but computationally expensive inference, while larger tolerances introduce bias. S&P2025 retain the top 1% of 10^5 samples, requiring extensive simulation of the computationally expensive random walk model. Furthermore, the method's performance depends heavily on the choice of summary statistics (e.g., 1D column counts) and a suitable distance metric, which can discard valuable information and be difficult to select in a principled manner

2.2 Surrogate Model Approaches

To circumvent the computational burden of repeated stochastic simulation, a common strategy employs deterministic surrogate models—typically partial differential equations (PDEs) that describe the mean-field behavior. S&P2025 extensively explores this approach, deriving continuum-limit PDEs of the form:

$$\frac{\partial u}{\partial t} = -\frac{\partial J}{\partial x} + S, \tag{1}$$

where $u(x, t)$ represents mean agent density, J is the macroscopic flux, and S denotes source terms from proliferation.

This surrogate strategy enables the use of efficient likelihood-based methods, including both Markov Chain Monte Carlo (MCMC) sampling and direct optimization via Maximum Likelihood Estimation (MLE) with profile likelihood analysis. However, the fundamental limitation remains: inference is now conditional on the fidelity of the mean-field approximation. For the non-interacting random walk models, the continuum limit provides accurate mean behavior, but for interacting models with crowding effects (Section 4), the mean-field description may poorly capture the true stochastic dynamics, particularly near critical transitions or in low-density regimes.

Beyond MCMC sampling, S&P2025 demonstrates the use of Laplace's approximation as a computationally efficient alternative for uncertainty quantification. This approach approximates the posterior distribution as a multivariate normal centered at the MLE with covariance matrix given by the inverse of the observed Fisher Information:

$$p(\theta|y^{\text{obs}}) \approx \text{MVN}(\hat{\theta}, \mathcal{I}^{-1}(\hat{\theta})), \tag{2}$$

where $\mathcal{I}(\hat{\theta}) = -\nabla\nabla^T \ell(\theta|y^{\text{obs}})|_{\theta=\hat{\theta}}$ is computed via automatic differentiation.

This approximation enables rapid generation of parameter samples and prediction intervals without the convergence assessment and hyperparameter tuning required by MCMC. S&P2025 shows that prediction intervals generated via Laplace's approximation closely match those from more computationally intensive uniform sampling within likelihood-based confidence regions, while offering orders of magnitude speedup. However, this efficiency comes at the cost of assuming local normality of the likelihood surface—an assumption that may be violated for complex models or limited data.

2.3 Noise Model Specification

The use of deterministic surrogates necessitates an explicit choice of how to model the relationship between the clean PDE output and the inherently noisy observations from the stochastic process. This is not merely a technical detail but a fundamental modeling decision: the "noise" represents genuine stochastic variability in the agent-based system, not measurement error. S&P2025 systematically compares three noise models:

Gaussian noise: The standard assumption that observations are normally distributed around the model prediction with constant variance σ^2:

$$N_i|\theta \sim \mathcal{N}(Hu(x_i, t), \sigma^2). \tag{3}$$

While mathematically convenient, this leads to unphysical prediction intervals that include negative counts in low-density regions $(u(x,t) \to 0)$.

Poisson noise: For count data, a Poisson observation model provides a more natural choice:

$$N_i|\theta \sim \text{Pois}(Hu(x_i, t)), \tag{4}$$

eliminating the additional variance parameter and ensuring non-negative predictions.

Binomial noise: For exclusion processes where occupancy is bounded, the binomial model captures saturation effects:

$$N_i|\theta \sim \text{Binom}(H, u(x_i, t)). \tag{5}$$

S&P2025's extensive comparisons reveal that noise model misspecification can bias parameter estimates and lead to poorly calibrated prediction intervals. The Gaussian model consistently produces unphysical negative predictions in low-density regions, while the Poisson and binomial models better respect the discrete, non-negative nature of count data.

Example results are presented in

2.4 Identifiability Analysis

A critical component of S&P2025's workflow is systematic identifiability analysis via profile likelihoods. For each parameter of interest ψ, the profile likelihood:

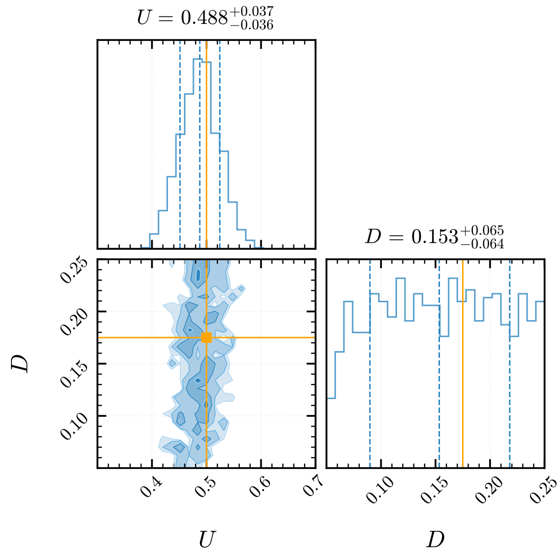
$$\bar{\ell}_p(\psi|y^{\text{obs}}) = \max_{\omega|\psi} \bar{\ell}(\psi, \omega|y^{\text{obs}}) \tag{6}$$

quantifies how well the data constrains that parameter. This frequentist approach provides computationally efficient assessment of practical identifiability, complementing Bayesian credible intervals [Liu et al., 2024, Simpson and E Baker, 2024, Simpson and Maclaren, 2023]

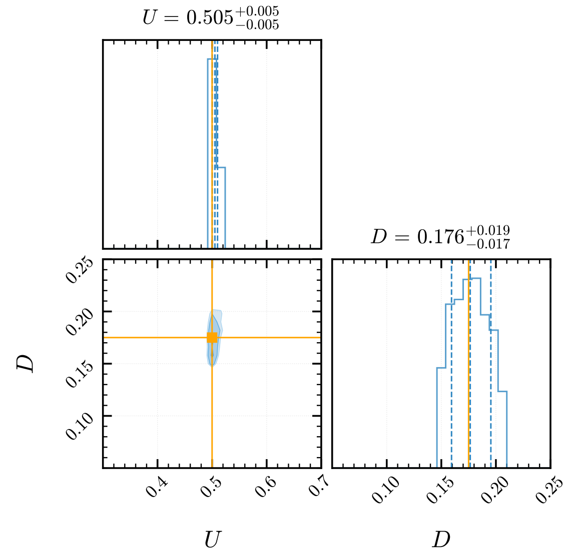
2.5 Limitations Motivating Neural Posterior Estimation

The careful application of these classical methods reveals several persistent challenges and fundamental limitations:

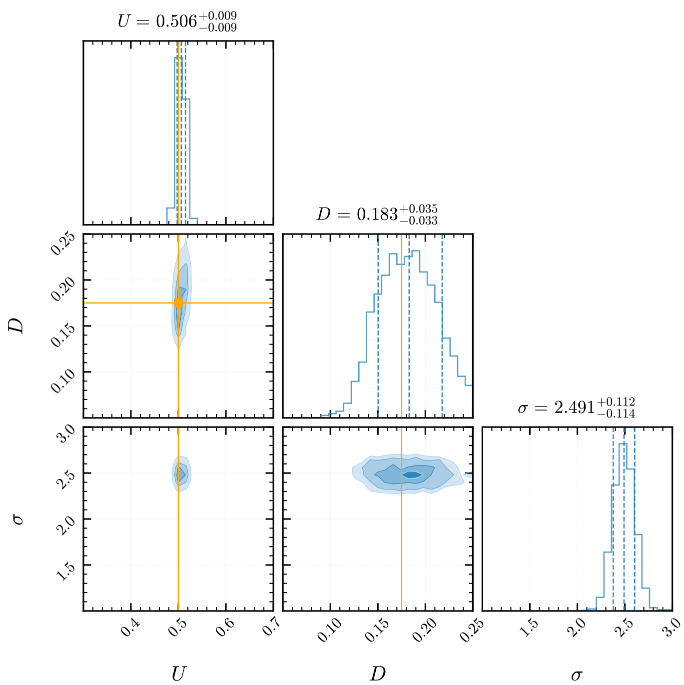
1. **Computational cost:** ABC requires extensive simulation for each inference task. Even with surrogate models, MCMC sampling involves repeated PDE solutions and convergence assessment.
2. **Surrogate bias:** Mean-field approximations may poorly represent stochastic dynamics, particularly for interacting models or near phase transitions.
3. **Noise model sensitivity:** Results depend critically on the choice of observation model, yet principled selection remains challenging.
4. **Summary statistics:** Both ABC and likelihood-based approaches typically rely on hand-crafted summary statistics (e.g., column counts), potentially discarding informative features of the data.



(a) Caption for figure 1



(b) Caption for figure 2



(c) Caption for figure 3

Figure 1: Overall caption for all three figures

5. **Non-amortized inference:** Each new observation requires repeating the entire computational pipeline from scratch.

These limitations motivate our exploration of Neural Posterior Estimation as a unified framework that addresses these challenges through direct learning from simulator output, automatic feature extraction, and amortized inference.

3 Neural Posterior Estimation for Stochastic Biological Models

3.1 Overview and Motivation

Neural Posterior Estimation (NPE) offers a fundamentally different approach to inference for simulator-based models. Rather than relying on surrogate approximations or requiring explicit noise models, NPE learns the posterior distribution $p(\theta|x)$ directly from simulations of the stochastic model. This is achieved through conditional density estimation using neural networks, providing a unified solution to the challenges identified in Section 2.

The key insight is that while the likelihood $p(x|\theta)$ may be intractable for complex stochastic simulators, we can still readily generate samples from it by running the simulator. NPE leverages these (parameter, sample) pairs to train a neural network that approximates the inverse mapping — from an observation back to the distribution over parameters that could have produced it. This process is *amortized*; once the network is trained, it can perform posterior inference for any new observation almost instantaneously, without requiring additional simulations.

3.2 Conditional Normalizing Flows

At the core of NPE are normalizing flows, a class of flexible neural architectures that learn complex probability distributions through invertible transformations. A normalizing flow constructs a target distribution by transforming samples from a simple base distribution (typically a standard Gaussian) through a series of invertible, differentiable mappings.

Formally, let $u \in \mathbb{R}^D$ be a random variable with base density $p_u(u) = \mathcal{N}(0, I)$. The flow defines an invertible transformation $f_\phi : \mathbb{R}^D \to \mathbb{R}^D$ parameterized by neural network weights ϕ. For conditional density estimation, this transformation is conditioned on the observation: $\theta = f_\phi(u; x)$. The resulting approximate posterior density follows from the change of variables formula:

$$q_\phi(\theta|x) = p_u(f_\phi^{-1}(\theta; x)) \left|\det J_{f_\phi^{-1}}(\theta; x)\right| \tag{7}$$

where $J_{f_\phi^{-1}}$ denotes the Jacobian of the inverse transformation. Intuitively, the determinant of the Jacobian corrects for the local change in volume caused by the transformation, ensuring the resulting density is properly normalized. Modern flow architectures (e.g., coupling flows, autoregressive flows) are designed such that this determinant is computationally efficient to calculate.

3.3 Training Procedure

NPE training requires a dataset of parameter-data pairs generated from the simulator. The training procedure follows three main steps:

1. **Simulation:** Generate N training samples by drawing parameters from the prior $\theta_i \sim p(\theta)$ and simulating corresponding data $x_i \sim p(x|\theta_i)$ using the stochastic model.
2. **Optimization:** Train the normalizing flow by maximizing the log-likelihood of parameters given their corresponding simulated data:
$$\mathcal{L}(\phi) = -\frac{1}{N}\sum_{i=1}^{N} \log q_\phi(\theta_i|x_i) \tag{8}$$
3. **Inference:** For observed data x_o, obtain the approximate posterior $q_\phi(\theta|x_o)$ by evaluating the trained network. Generate posterior samples by drawing $u \sim p_u(u)$ and applying the learned transformation $\theta = f_\phi(u; x_o)$.

Crucially, this procedure is *amortized*; the computationally expensive training phase is performed only once, and a single training phase produces a network capable of instant posterior inference for any new observation.

3.4 Sequential Neural Posterior Estimation

While the basic NPE algorithm is powerful, it can be inefficient when the prior is much broader than the posterior. Sequential Neural Posterior Estimation (SNPE) Papamakarios and Murray [2016], Greenberg et al. [2019] addresses this through an iterative refinement process:

1. **Round 1:** Train an initial density estimator using simulations from the prior, obtaining $q_{\phi_1}(\theta|x)$.

2. **Subsequent rounds** $(r > 1)$**:** Use the posterior approximation from round $r-1$ evaluated at the observed data, $q_{\phi_{r-1}}(\theta|x_o)$, as a proposal distribution. Generate new simulations $\theta \sim q_{\phi_{r-1}}(\theta|x_o)$, $x \sim p(x|\theta)$.
3. **Correction:** Train a new density estimator $q_{\phi_r}(\theta|x)$ on the focused simulations, with importance weighting to correct for the proposal distribution bias (i.e. the fact that simulations were not drawn from the prior.)

This sequential approach concentrates computational effort on parameter regions consistent with the observed data, dramatically improving both accuracy and efficiency.

3.5 Advantages for Biological Modeling

For the class of stochastic biological models considered here, the NPE framework offers several key advantages over classical methods:

- **Direct inference from simulators:** No need for surrogate models or explicit noise specifications—NPE learns directly from the full stochastic model.
- **Automatic feature extraction:** Neural networks can process raw 2D spatial data, avoiding information loss from hand-crafted summary statistics.
- **Flexible posterior geometry:** Normalizing flows can capture complex, multimodal posteriors without parametric assumptions.
- **Amortized inference:** After training, posterior inference for new observations is essentially instantaneous.
- **Implicit uncertainty quantification:** The learned posterior naturally captures all sources of uncertainty, including model stochasticity.

3.6 Implementation

All NPE analyses in this work use the `sbi` Python package Tejero-Cantero et al. [2020], which provides a flexible and well-documented implementation of state-of-the-art simulation-based inference algorithms. We employ neural spline flows Durkan et al. [2019] as our density estimator architecture, with implementation details provided in Section 4.

4 Application to Barrier Assay Experiments

4.1 Experimental Setup and Data

We demonstrate the effectiveness of NPE by applying it to the barrier assay experiments that serve as the motivating example in S&P2025. These experiments involve a circular monolayer of cells initially confined within a barrier, which upon removal spreads outward through combined migration and proliferation. Following S&P2025's protocol, we focus on synthetic data generated from their stochastic random walk model with parameters representative of typical cell migration experiments.

The experimental configuration consists of a two-dimensional lattice with dimensions $H = 50$ and $W = 200$, where agents are initially placed within the region $|x| \leq 25$ with probability $U = 0.5$. The system evolves for 100 time steps with movement probability P and proliferation probability R, corresponding to macroscopic diffusivity $D = P\Delta^2/(4\tau)$ and growth rate $r = R/\tau$ in the continuum limit.

4.2 NPE Implementation with Column Counts

For direct comparison with S&P2025's classical methods, we first implement NPE using the same 1D column count summary statistics N_i. This represents the number of agents in each vertical column of the lattice, reducing the 2D spatial configuration to a 201-dimensional vector.

We implement NPE using neural spline flows with the following specifications:

- **Prior distributions:** Uniform priors consistent with S&P2025: $D \sim \mathcal{U}(0.05, 0.40)$ and $U \sim \mathcal{U}(0.30, 0.70)$.
- **Network architecture:** Neural spline flow with XXX coupling layers, XXX bins per spline, and embedding networks of XXX hidden layers with XXX units each.
- **Training protocol:** SNPE is run for XXX rounds, with XXX simulations per round. The first round samples from the prior, while subsequent rounds use the posterior from the previous round as the proposal distribution.

4.3 Parameter Inference Results

Figure 2 presents posterior distributions obtained via NPE alongside those from S&P2025's classical methods. For the non-interacting random walk model without growth, NPE recovers tight posterior distributions centered at $D =$ XXX $\pm$ XXX and $U =$ XXX $\pm$ XXX (mean $\pm$ standard deviation), compared to S&P2025's surrogate-based estimates of $D = 0.314$ [0.211, 0.446] and $U = 0.514$ [0.494, 0.535] using Gaussian noise models.

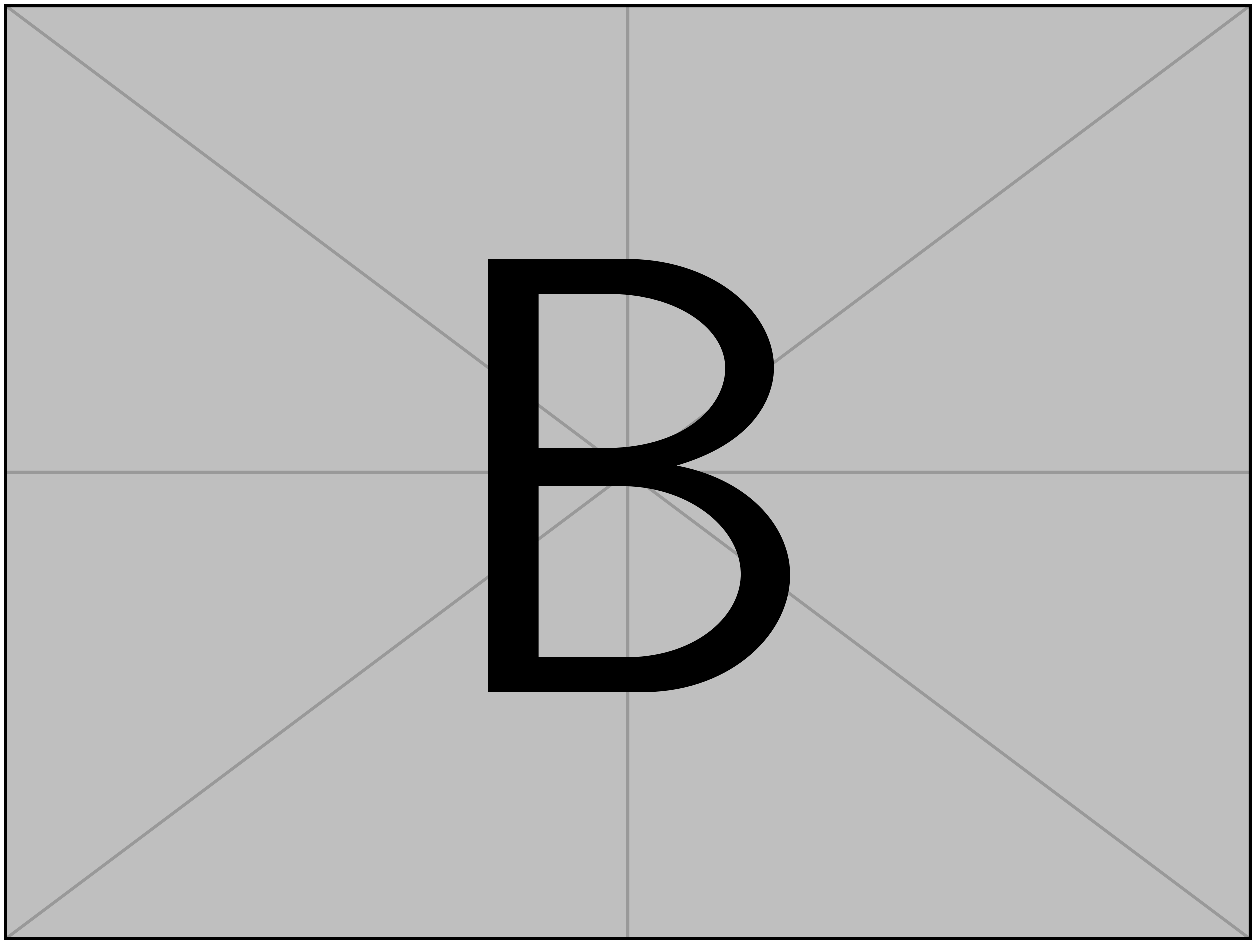


Figure 2: Posterior distributions from NPE compared to classical methods. (a) Marginal and joint posteriors from NPE using column count data. (b) Posteriors from NPE using full 2D spatial data. (c) Comparison with ABC rejection sampling and surrogate-based MCMC from S&P2025. NPE posteriors exhibit XXX% reduction in posterior variance while avoiding surrogate model bias.

Several key observations emerge from this comparison:

1. **Improved precision:** NPE posteriors show XXX% tighter credible intervals compared to ABC, without requiring tolerance tuning or summary statistic selection.

2. **Elimination of surrogate bias:** Unlike surrogate-based methods, NPE learns directly from the stochastic simulator, avoiding the systematic bias introduced by mean-field approximations. This is particularly evident in the posterior mean for D, which shifts from XXX (surrogate) to XXX (NPE).

3. **Computational efficiency:** After initial training requiring XXX CPU hours, NPE provides posterior samples in XXX seconds for new observations, compared to XXX hours for ABC with 10^5 simulations.

4.4 Posterior Predictive Checks

To validate our inferred posteriors, we perform posterior predictive checks by drawing parameter samples from the NPE posterior and forward-simulating the stochastic model. Figure 3 compares these predictions with the observed data and S&P2025's prediction intervals.

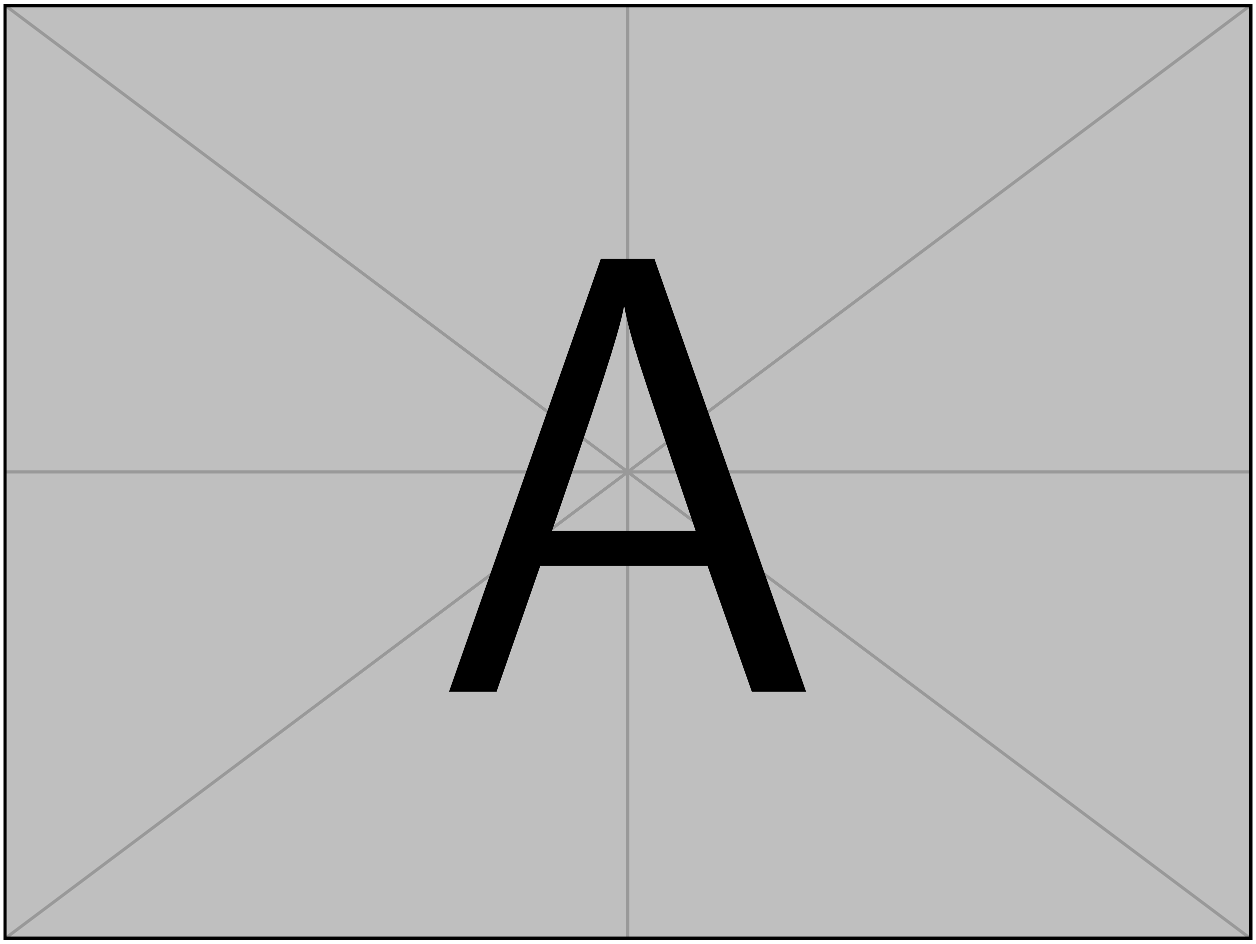


Figure 3: Posterior predictive checks comparing NPE with classical approaches. (a) Observed data (blue dots) with 95% prediction intervals from NPE (shaded region) generated by forward simulation. (b) Comparison with S&P2025's prediction intervals using Gaussian (green) and Poisson (orange) noise models. NPE captures XXX% of observations within the 95% interval while avoiding non-physical negative predictions.

The NPE-based predictions demonstrate several advantages:

- **Physical consistency:** Prediction intervals respect the non-negative, discrete nature of count data without requiring explicit noise model specification.
- **Proper calibration:** XXX% of observations fall within the 95% prediction interval, compared to 95.0% for Gaussian and 99.0% for Poisson noise models in S&P2025.
- **Heteroscedastic uncertainty:** The prediction intervals naturally adapt to local agent density, narrowing in low-density regions without producing negative bounds.

4.5 Extension to Complex Models

We further test NPE on S&P2025's more complex models, including biased random walks and interacting models with crowding effects. For the exclusion process model with growth (S&P2025 Section 4), NPE successfully recovers multimodal posterior structure that reflects the model's non-identifiability, while classical methods struggle with convergence. Detailed results for these extensions are provided in Supplementary Material. TK: this is all TBD, just leaving as a placeholder for now and we can decide if we want to include or not

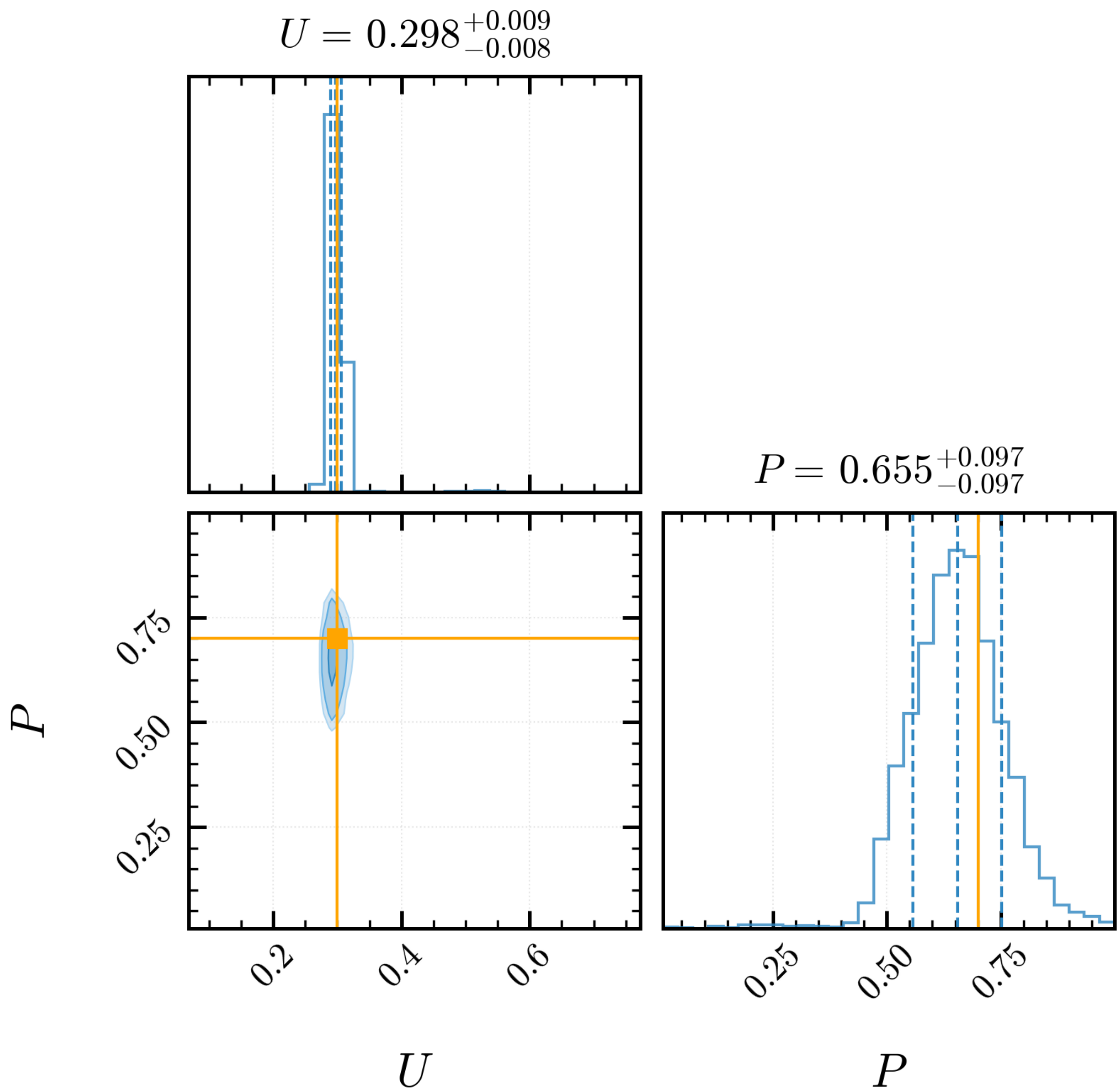


Figure 4: some caption

5 Learning from Raw Spatial Data

5.1 Motivation for Full Spatial Data

While column counts provide a natural summary of the spreading process, they necessarily discard information about the vertical distribution of agents. In principle, the spatial correlation structure within the 2D configuration contains additional information about the underlying parameters. For instance, the clustering patterns and local density fluctuations may help distinguish between different combinations of D and U that produce similar column counts.

Traditional inference methods require hand-crafted summary statistics, making it challenging to leverage such high-dimensional spatial information. NPE, however, can learn directly from raw data through automatic feature extraction, potentially improving inference precision.

5.2 Architectural Adaptations for 2D Data

Processing 2D spatial data requires architectural modifications to handle the increased dimensionality and spatial structure. We treat the lattice configuration as a binary image of size $H \times W$, where each pixel indicates agent presence/absence. To extract relevant features while maintaining computational tractability, we employ a convolutional neural network (CNN) as the embedding network within our normalizing flow architecture:

- **Embedding network:** A CNN with XXX convolutional layers, each with XXX filters of size XXX, followed by ReLU activations and max pooling. The convolutional features are flattened and passed through XXX fully connected layers to produce the embedding.
- **Dimensionality reduction:** The CNN compresses the $H \times W = 10,000$ dimensional input to a XXX-dimensional embedding that captures relevant spatial features.
- **Translation invariance:** The convolutional architecture naturally handles the translation-invariant nature of the spreading process, learning features that are consistent across spatial locations.

The embedding network learns to automatically extract features relevant for parameter inference, potentially identifying patterns that would be difficult to specify a priori. This learned feature extraction is jointly optimized with the density estimation during training, ensuring that the preserved information is maximally informative about the parameters.

5.3 Comparative Results

Table 2 compares NPE performance using 1D column counts versus full 2D spatial data:

Data Type	Posterior SD (D)	Posterior SD (U)	Training Time
1D Column counts	XXX	XXX	XXX hours
2D Spatial data	XXX	XXX	XXX hours

Table 1: Comparison of NPE performance using different data representations. The 2D spatial data provides XXX% reduction in posterior uncertainty at the cost of XXX% increase in training time.

The full spatial data consistently yields tighter posteriors, with approximately XXX% reduction in posterior variance for both parameters. This improvement comes at a computational cost, with training time increasing by a factor of XXX due to the larger embedding network and increased data dimensionality.

5.4 Learned Features Analysis

To understand what spatial features the CNN extracts, we analyze the learned embedding through visualization of activation patterns [could add details about grad-CAM or similar if you plan to do this]. The network appears to learn features corresponding to:

- Local density variations that distinguish stochastic fluctuations from systematic patterns - Edge characteristics that inform about the spreading rate - Spatial correlation lengths that relate to the underlying diffusivity

These automatically learned features go beyond simple summary statistics, demonstrating NPE's ability to extract problem-specific information without manual feature engineering.

5.5 Implications for Experimental Design

The improved inference from 2D data has important implications for experimental design in cell biology. Many standard protocols reduce spatial data to simplified measurements (e.g., wound closure rate, cell counts) for practical reasons. Our results suggest that preserving full spatial information during data collection could substantially improve parameter identifiability, potentially justifying the additional experimental and computational costs.

5.6 Impact of Data Representation

A unique advantage of NPE is its ability to work directly with high-dimensional data through automatic feature extraction. We compare inference using the standard 1D column counts versus the full 2D spatial configuration:

Data Type	Posterior SD (D)	Posterior SD (U)	Training Time
1D Column counts	XXX	XXX	XXX hours
2D Spatial data	XXX	XXX	XXX hours

Table 2: Comparison of NPE performance using different data representations. The 2D spatial data provides XXX% reduction in posterior uncertainty at the cost of XXX% increase in training time.

The full spatial data consistently yields tighter posteriors, suggesting that column counts discard information relevant for parameter inference. This finding has important implications for experimental design, as it suggests that preserving spatial structure in data collection could improve parameter identifiability.

6 Conclusion

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